

COLEGIO DE MONTALBAN

CDM Library Mobile App - Promotional Portal

Technical Web Documentation & Panel Presentation Guide

Target: Promotional Web Landing Page

Stack: HTML5, Vanilla CSS3, JavaScript ES6+

Integrations: REST API & SQLite

1. Executive Summary

The **CDM Library Mobile App Promotional Portal** is a high-performance web landing page designed to promote the mobile application to students of **Colegio de Montalban (CDM)**. It provides direct APK downloads, side-loading installation instructions, real-time live catalog search, and architectural visualization of student-to-librarian data synchronization.

2. Technology Stack & Design Tokens

Layer	Technologies	Purpose & Implementation
Structure	HTML5	Semantic document layout with responsive containers, sections, and accessibility tags.
Styling	Vanilla CSS3	Custom CSS design system featuring Flexbox, CSS Grid, custom properties (variables), and smooth transitions.
Scripting	JavaScript (ES6+)	Asynchronous API data fetching (<code>fetch()</code>), live search indexing, phone mockup rendering, and accordion triggers.
Integration	Express REST API	Queries <code>http://localhost:5002/api/books</code> with an offline fallback dataset (<code>MOCK_BOOKS</code>) for network resilience.

3. Core Web Modules

- Hero & Interactive Phone UI:** Features the "Official Student Portal" tagline, quick KPI stats, and an interactive smartphone visualizer that live-syncs search queries with the main page.
- Features & Architecture Sync Diagram:** Demonstrates real-time data synchronization between student mobile devices and the librarian's desktop Express backend.
- Live Catalog Search Engine:** Allows students to search titles, authors, and categories in real-time without logging in, displaying availability status (`Available` vs `Borrowed`).

- **Direct APK Download Zone:** Direct installer download link (`cdm-library-app.apk`) with iOS roadmap status.
- **3-Step Installation Guide:** Collapsible instructions for downloading the APK, enabling "Install Unknown Apps" in Android security settings, and running the package installer.

4. Capstone Attribution (BSIT 4C - Group 9)

Developed by Group 9 • BSIT 4C

De Guzman, Ronjay E.: Project Leader & Programmer

Martinez, Cristan Irex: Programmer

Lirasan, Georgette Mae P.: Manuscript Writer

Remulta, Bryan P.: Diagram Designer

5. Defense Panel FAQ Cheatsheet

Q1: Why build a promotional website alongside the desktop and mobile apps?

Since the Android APK is distributed directly for campus use rather than published on public commercial app stores, the promotional portal acts as the official, secure campus distribution hub for downloads and installation guides.

Q2: Does searching books on the web portal require a student account?

No. Open search on the web landing page allows students to quickly check book availability on campus without logging in. Account authentication is only required for personal borrowing and reservations.

Q3: How does the web portal handle backend server connection failures?

The website includes a built-in offline fallback mechanism. If the Express server is offline, JavaScript automatically catches the network error and populates the catalog search using a local mock dataset.